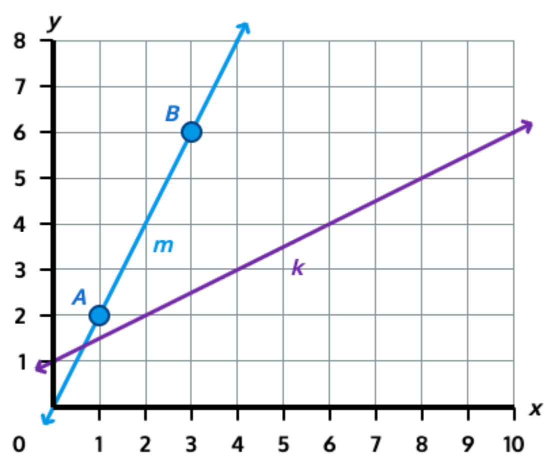
Mini Lesson Sample Script

Materials: Straightedge

<table>
<tr><td colspan="2">1. Compare the steepness of two lines on a grid and find the slope of one line using a slope triangle.</td></tr>
<tr><td>Which line is steeper, line m or line k?

Then find the slope of line m. Use points A and B on line m to help.



Answer: Line m is steeper; the slope of line m is 2.</td><td>T: Picture each line as a slide. If you slid down each one, which would you go down faster?

S: Line m. It leans more.

T: Right, so line m is steeper than line k. We can describe a line's steepness with a number called the slope. To find it, we draw a slope triangle between two points and divide the vertical distance by the horizontal distance. Let's use points A and B on line m. I'll draw the vertical side going up from A, then the horizontal side over to B. How many units is the vertical side?

S: 4.

T: And the horizontal side?

S: 2.

T: So the slope is vertical over horizontal. What's 4 divided by 2?</td></tr>
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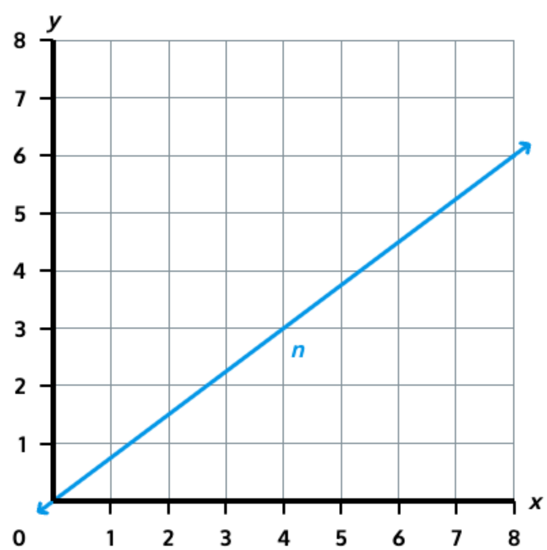
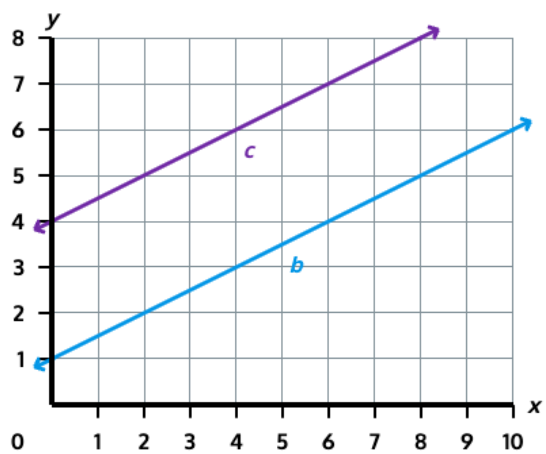
	S: 2. T: The slope of line m is 2. And does it make sense that line m has the larger slope? S: Yes, because it's the steeper line. A bigger slope means steeper.
2. **Find the slope of a line on a grid by drawing a slope triangle between two points.**	
What is the slope of line n? Choose two points on the line where grid lines intersect. Draw a slope triangle between them, then find the slope.  *Answer: The slope of line n is 3/4. (Any two lattice points on line n produce the same slope because all slope triangles on a line are similar.)*	T: For this one, pick two points on line n where the line crosses a grid intersection. Why are those points helpful? S: Because I can count the units exactly. T: Go ahead and mark two points and draw the slope triangle between them with your straightedge. S: (Marks points at (0, 0) and (4, 3). Draws horizontal and vertical sides.) T: How long is the horizontal side, and how long is the vertical side? S: The horizontal side is 4 and the vertical side is 3. T: So what's the slope?

	S: 3 over 4. T: If you had picked two different lattice points on line n, what would you have gotten? S: The same thing, 3/4. The slope triangles would still be similar.
3. **Find the slope of a line parallel to a line with a known slope.**	
Lines b and c are parallel. The slope of line b is 1/2. What is the slope of line c?  *Answer: The slope of line c is 1/2.*	T: Remember, parallel lines have the same slope. Find the slope of line c on your own. S: (Works.)